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COURSE 1.14 / COMPANION WORKBOOK

Deliver Windows updates from SBD-WSUS01

Air-gapped WSUS: offline import, four rings & Group Policy targeting

Give every Windows system one road for updates: build SBD-WSUS01 as a disconnected WSUS server, feed it each month through the export and import path Microsoft describes for air-gapped networks, target four rings by Group Policy — pilot, workstations, member servers, then the domain controllers and the rest of Tier 0, installed by hand — and prove the cycle with WSUS reports, spot checks and a first STIG baseline.

MATERIALS EDITION 2.0 / COURSE 1.14

Updated 14 October 2026 · sbd-patch-protect

NAME _____ / DATE _____

START HERE / YOUR LEARNING PATH

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

Before you start

This course gives every Windows system one road for updates. SBD-WSUS01 never touches the internet: each month a WSUS server on the connected side synchronizes the same scope, and its metadata and update files cross on hashed media to be imported inside. Microsoft deprecated WSUS in September 2024 — it gets no new features — but it still ships in Windows Server and is supported for production use, with security and quality updates under the product lifecycle. Ring timings and dates are sample values for your patch owner to set. Linux packages already have their road, SBD-REPO01 (course 1.9); endpoint protection and credentialed scanning are coming soon, as courses 1.16 and 1.17, and nothing here depends on them.

Prerequisites

- ☐ Complete courses 1.1–1.13: the domain with its DNS and time design, the STIG baselines and hardening policies, the PKI, DHCP, the tiered administration seats, SBD-REPO01, the logging pipeline with SBD-AUDIT01, run by its audit custodian (course 1.11), and the fleet of SBD-PC01–SBD-PC50 with its three pilot machines.
- ☐ Provision SBD-WSUS01 (10.0.20.30, VLAN 20) as a Windows Server virtual machine with a separate data volume (D: in this sample) sized for the update files.
- ☐ Outside the boundary, have an approved connected system that can run a second WSUS server with access to Microsoft Update, and the course 1.4 removable-media procedure for carrying its output across.

What this course builds

- Build SBD-WSUS01 as a disconnected update server with TLS-protected metadata and tier-bound administration.
- Carry each month's updates across the air gap by WSUS export and import, hashed on both sides.
- Target four update rings with computer groups and Group Policy, with the domain controllers installed last and by hand.
- Promote a monthly cycle ring by ring and prove it with compliance reports, spot checks and a first baseline.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-WSUS01 — name, address, OS build	

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LESSON 01 / ABOUT 25 MIN

Design the rings and the Windows update rule

WHY THIS MATTERS Decide how Windows updates reach the network before any arrive: one road through SBD-WSUS01, four rings that try every update on what the network can lose before what it can't, and a monthly crossing on the media procedure. Then confirm the register already carries the road.

In this lesson you will

- Write the Windows update rule and its documented exceptions.
- Place every Windows system in one of four rings, in order.
- Set the timing and the crossing cadence, and confirm the register rows.

Follow the build

1. Write the Windows update rule — the network's one-source-per-type rule, applied to Windows: every Windows system takes updates only from SBD-WSUS01, and anything else is a documented exception. SBD-ROOTCA01 is the standing one: it has no network, so any update it takes is carried by hand at its next ceremony and recorded. Linux packages already have their one source, SBD-REPO01 (course 1.9); protection and scanner content are outside this course.
2. Place every Windows system in one of four rings, in the order rule F06 names. Ring 1, the pilot: the three course 1.13 pilot PCs (SBD-PC01–SBD-PC03) and SBD-WSUS01 itself, the first server to take each server update and one whose loss stops only patching. Ring 2: the other 47 workstations. Ring 3: the member servers and the Tier 1 and Tier 2 seats — SBD-ADMIN-SRV01 and its SBD-ADMIN-PC01 virtual machine — so the seats that would fix a broken workstation update still run the proven one. SBD-AUDIT01 takes Ring 3's updates at Ring 3's time but in a computer group of its own, Ring 3 - Audit archive: they download only, and its audit custodian installs them, Ring 4-style, because the custodian alone administers that server (course 1.11) — never ordinary Tier 1 operators.
3. Ring 4 holds the domain controllers and the rest of Tier 0: SBD-SUBCA01, SBD-PKI-MGMT01 and AdminJumpBox01. It goes last and is Tier 0 work, approved and installed from AdminJumpBox01 with Tier 0 accounts, one system at a time. SBD-WSUS01 itself stays a Tier 1 member server run by the patch operators: Tier 0's hold on Ring 4 is a written procedure (lesson 2), and moving SBD-WSUS01 to Tier 0 is recorded as the target.
4. Write the timing as sample values your patch owner approves: a ring advances only after its machines have installed and reported cleanly for a set soak time — three days after Ring 1 and five after Rings 2 and 3 in this sample. A failed or harmful update stops promotion until it is understood, and nothing skips a ring.
5. Set the crossing cadence: one crossing a month, after Microsoft's monthly security release on the second Tuesday, through the course 1.4 media procedure and on the same run as SBD-REPO01's monthly snapshot (course 1.9) — one custody record, one approval. An urgent out-of-band fix gets an expedited crossing on the same procedure, with shorter soak times the owner approves in writing.
6. Confirm the register: F06 already lets every Windows system reach SBD-WSUS01 — metadata over HTTPS 8531, signed content over HTTP 8530 — with the patch owner as its owner, and A01 and A02 carry the two consoles. Nothing leads out of the network and no row is added for one: the content arrives on media.

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Check your result

```
# On SBD-ADMIN-SRV01: every Windows computer object and its OU, the input for the ring
table
New-Item -ItemType Directory -Force C:\Evidence\patch-protect | Out-Null
([adsisearcher]'(&(objectCategory=computer)(operatingSystem=Windows*))').FindAll() |
ForEach-Object { $_.Properties.distinguishedname[0] } | Sort-Object | Set-Content C:\
Evidence\patch-protect\windows-computers.txt
(Get-Content C:\Evidence\patch-protect\windows-computers.txt).Count
# On any member: no update policy yet, so no WSUS server is named (expected: no output)
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate' -ErrorAction
SilentlyContinue
```

WHAT SUCCESS LOOKS LIKE The rule names its one source and its exceptions, the ring table places every Windows computer object in exactly one ring with the reason for its place, SBD-AUDIT01 in its custodian-installed group, the timing and crossing calendar are written with an approver, and F06, A01 and A02 are confirmed with owners.

Your record

Evidence to keep: Keep the Windows update rule, the computer list and the ring table built from it, the timing and crossing calendar, and the confirmed register rows.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why do domain controllers sit in the last ring?

REMEMBER One road: Windows updates only from SBD-WSUS01, exceptions written down. Four rings, pilot to Tier 0 — updates earn their way up. One monthly crossing beside SBD-REPO01's, with an expedited path.

Microsoft · Windows Server Update Services (WSUS) overview

Microsoft · Plan your WSUS deployment

Microsoft · Deploy Windows client updates using WSUS (rings and client-side targeting)

NIST · SP 800-53 Rev. 5 security controls

LESSON 02 / ABOUT 40 MIN

Build SBD-WSUS01 as a disconnected server

WHY THIS MATTERS Build the server the network will take its updates from: a hardened member server on VLAN 20 that never synchronizes, serves metadata over TLS and signed files over HTTP, and is administered only from the two seats the tier model allows. Its scope is decided here, because what crosses each month is whatever that scope holds.

In this lesson you will

- Install WSUS on a hardened member server configured never to synchronize.
- Decide and record the update scope the monthly crossing will carry.
- Protect its metadata with TLS and its administration with the tier model.

Follow the build

1. Build SBD-WSUS01 as a Windows Server virtual machine at 10.0.20.30 on VLAN 20, with a separate data volume for the update files (D: in this sample), both domain controllers for DNS, and the hypervisor's time synchronization turned off, so it takes time from the domain hierarchy (course 1.3). Join it straight into the Servers OU, so course 1.5's member-server STIG policies, course 1.8's tier logon policies and course 1.10's enrolling GPO, linked to the Servers OU when course 1.10 enrolled SBD-DHCP01, apply from its first boot; confirm with w32tm /query /source that it takes time from a domain controller.
2. Install the role with its default database, the Windows Internal Database (WID), and run the post-installation step with the content folder on D:. Microsoft announced in September 2024 that WSUS is deprecated: it gets no new features, but it still ships in Windows Server and is supported for production use, with security and quality updates under the product lifecycle. WID is deprecated too — Microsoft suggests SQL Server for the roles that use it — so record the database choice as a decision to revisit before either is removed.
3. Configure it as a disconnected server: cancel the configuration wizard, which needs Microsoft Update to list products, then in Options store update files locally, enable only the languages your systems use, leave express installation files off (the default), and keep synchronization manual — and never run it. SBD-WSUS01 has no route out; its updates arrive only by import (lesson 3).
4. Decide the scope and record it: the Windows Server and Windows client products the network runs, security and critical updates and — if Microsoft Defender Antivirus protects the endpoints — its definition updates; no upgrades and no drivers, which are planned changes of their own. The connected-side server in lesson 3 synchronizes exactly this scope, because whatever it holds is what crosses.
5. Protect the metadata: add SBD-WSUS01 to the SBD TLS Servers group, then request an SBD Server Authentication certificate (course 1.6) for SBD-WSUS01.sbd.test, bind it to the WSUS Administration site's HTTPS port 8531, require TLS on the five web services Microsoft lists (APIRemoting30, ClientWebService, DSSAuthWebService, ServerSyncWebService and SimpleAuthWebService), and run wsusutil configuressl. Update files stay on HTTP 8530, as F06 says: each one is signed, and its hash reaches the client inside the TLS-protected metadata, so a changed file is refused.
6. Hold the administration to the tier model: scope the 8530 and 8531 inbound rules to the VLANs that hold F06's Windows systems, install the WSUS console on SBD-ADMIN-SRV01 and AdminJumpBox01, and keep WSUS Administrators to the named patch operators and the Tier 0 task accounts that run Ring 4.

From AdminJumpBox01, add SBD-WSUS01 to SBD Tier 0 Console Hosts (course 1.8's recorded network-logon exception for this console) and to SBD Event Sources (course 1.10), and restart it once the collector's WinRM scope includes 10.0.20.30 (F17). WSUS can't limit an administrator to some computer groups, so Ring 4's Tier 0 rule — like the audit custodian's separation for SBD-AUDIT01 — is a written procedure, not a technical boundary. Write the Ring 4 procedure — Ring 4 is approved and installed only by Tier 0 accounts from AdminJumpBox01, and every Ring 4 approval is checked by name before a controller installs (lesson 5) — and record in the exception record the residual risk that a Tier 1 operator could still approve for Ring 4, with the target that closes it: moving SBD-WSUS01 to Tier 0, administered only from AdminJumpBox01. Until then SBD-WSUS01 stays Tier 1.

Check your result

```
w32tm /query /source   # a domain controller, never the hypervisor's time provider
Install-WindowsFeature -Name UpdateServices -IncludeManagementTools
& "$env:ProgramFiles\Update Services\Tools\wsusutil.exe" postinstall CONTENT_DIR=D:\WSUS
$wsus = Get-WsusServer
$wsus.GetConfiguration() | Select-Object HostBinariesOnMicrosoftUpdate,
DownloadExpressPackages, AllUpdateLanguagesEnabled   # expected after Options: False,
False, False
$wsus.GetSubscription() | Select-Object SynchronizeAutomatically   # expected: False
# TLS for metadata: the certificate, the 8531 binding, the five services, then wsusutil
$cert = (Get-Certificate -Template SBDServerAuthentication -SubjectName 'CN=SBD-
WSUS01.sbd.test' -DnsName SBD-WSUS01.sbd.test -CertStoreLocation Cert:\LocalMachine\
My).Certificate
(Get-WebBinding -Name 'WSUS Administration' -Protocol
https).AddSslCertificate($cert.Thumbprint, 'My')
'APIRemoting30', 'ClientWebService', 'DSSAuthWebService', 'ServerSyncWebService',
'SimpleAuthWebService' | ForEach-Object { Set-WebConfigurationProperty -PSPath 'IIS:\' -
Location "WSUS Administration/$_" -Filter system.webServer/security/access -Name sslFlags
-Value Ssl }
& "$env:ProgramFiles\Update Services\Tools\wsusutil.exe" configuressl SBD-WSUS01.sbd.test
Get-WebBinding -Name 'WSUS Administration' | Select-Object protocol, bindingInformation
# http *:8530, https *:8531
Get-LocalGroupMember 'WSUS Administrators'
```

> **WHAT SUCCESS LOOKS LIKE** SBD-WSUS01 sits in the Servers OU, takes time from a domain controller, and runs WSUS on WID, stores files locally, never synchronizes, and answers metadata over TLS on 8531 with a certificate that chains to the SBD root; WSUS Administrators holds only the named operators and Tier 0 task accounts; the server is in SBD Tier 0 Console Hosts and forwards its events; and the scope, the database decision, the written Ring 4 procedure, and its residual risk with the Tier 0 target are recorded.

Your record

Evidence to keep: Keep the installation and configuration output, the scope and database decisions, the certificate and binding details, the WSUS Administrators list, the group memberships, the written Ring 4 procedure, and the exception-record entry for the residual risk and its Tier 0 target.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

How does update content reach this WSUS server?

> **REMEMBER** A hardened member server that never synchronizes. Metadata over TLS on 8531; signed files over HTTP on 8530. Two consoles, named administrators, and a written Ring 4 procedure for the limit WSUS can't enforce, with Tier 0 as the recorded target.

Microsoft · Windows Server Update Services (WSUS) overview

Microsoft · Features removed or no longer developed in Windows Server

Microsoft · Plan your WSUS deployment

Microsoft · Step 1: Install the WSUS server role

Microsoft · Step 2: Configure WSUS (TLS, computer groups, air-gapped networks)

NIST · SP 800-53 Rev. 5 security controls

LESSON 03 / ABOUT 35 MIN

Carry updates across the air gap

> **WHY THIS MATTERS** SBD-WSUS01 never talks to Microsoft. Each month a second WSUS server on the connected side synchronizes the recorded scope and downloads its files; its metadata export and content folder cross on approved media, hashed on both sides, and SBD-WSUS01 imports them — the two-server path Microsoft describes for air-gapped networks.

In this lesson you will

- Run the connected-side export server with SBD-WSUS01's recorded scope.
- Export the metadata and the files, and carry both under the media procedure.
- Verify the crossing on SBD-WSUS01 before importing, and file its record.

Follow the build

1. On the approved connected system, run the export server: WSUS on the same Windows Server release as SBD-WSUS01, synchronizing from Microsoft Update with exactly the products, classifications and languages lesson 2 recorded, and storing files locally. It sits outside the boundary and never connects to the lab; its only output is media.
2. After each month's synchronization, approve the updates you need on the export server so it downloads their files, and wait until nothing is still downloading — those approvals only switch the downloads on, and they stay outside. Then, as a local administrator on the export server, run wsusutil export for the metadata package and copy the WsusContent folder: the package carries the update metadata, and the files travel beside it (after the first crossing, only those added since the last one).
3. Hash the package and every file into connected-side-hashes.csv, which travels with the media, and carry the set under the course 1.4 media procedure with its custody record, on the run that also carries SBD-REPO01's snapshot. Declining an update doesn't stop it from being exported, so the export server's scope is the real limit on what crosses.
4. On SBD-WSUS01, hash the media's contents into lab-side-hashes.csv and compare the two lists before opening anything: one mismatch stops the import and becomes a finding. Copy the files into D:\WSUS\WsusContent first, then run wsusutil import locally as a local administrator — Microsoft warns never to import an export from a source you don't trust, and the custody record and matching hashes are how this network earns that trust. A large import can run for hours.
5. Check the import log, confirm the month's updates now appear in the console, and file the crossing in C:\Evidence\patch-protect\transfer: dates, carrier, file count, the package hash, both hash lists and the import log. If WSUS later shows an approved update as still waiting for its files, wsusutil reset makes it re-check its content folder against the database.

Check your result

```
# Connected side, on the export server, once every needed file has downloaded
& "$env:ProgramFiles\Update Services\Tools\wsusutil.exe" export E:\2026-10\wsus-export.xml.gz E:\2026-10\export.log
robocopy D:\WSUS\WsusContent E:\2026-10\WsusContent /E   # the first crossing carries the whole folder
Get-ChildItem E:\2026-10 -Recurse -File | Get-FileHash -Algorithm SHA256 | Export-Csv E:\connected-side-hashes.csv -NoTypeInformation
```

```
# Inside, on SBD-WSUS01, with the media as E: (no output from Compare-Object means every
hash matches)
$t = 'C:\Evidence\patch-protect\transfer\2026-10'; New-Item -ItemType Directory -Force $t
| Out-Null
Get-ChildItem E:\2026-10 -Recurse -File | Get-FileHash -Algorithm SHA256 | Export-Csv "$t\
lab-side-hashes.csv" -NoTypeInformation
Compare-Object (Import-Csv E:\connected-side-hashes.csv).Hash (Import-Csv "$t\lab-side-
hashes.csv").Hash
robocopy E:\2026-10\WsusContent D:\WSUS\WsusContent /E
& "$env:ProgramFiles\Update Services\Tools\wsusutil.exe" import E:\2026-10\wsus-
export.xml.gz "$t\import.log"
Copy-Item E:\connected-side-hashes.csv, E:\2026-10\export.log -Destination $t
```

> **WHAT SUCCESS LOOKS LIKE** The export server's scope matches lesson 2's record, every hash matches on both sides, the import log reports success, the month's updates appear on SBD-WSUS01, and the crossing record is complete.

Your record

Evidence to keep: Keep the export server's scope settings, both hash lists and their comparison, the custody record, the export and import logs, and the crossing record with the package hash and file count.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why must the export server synchronize exactly the recorded scope?

> **REMEMBER** Two WSUS servers: one outside fills the media, one inside serves the network. Package and files cross hashed, under custody. Compare first, copy the files, then import — or stop.

Microsoft · Plan your WSUS deployment

Microsoft · Step 2: Configure WSUS (TLS, computer groups, air-gapped networks)

Microsoft · Synchronize software updates from a disconnected software update point (WSUSUtil export and import)

Microsoft · Get-FileHash

NIST · SP 800-53 Rev. 5 security controls

LESSON 04 / ABOUT 40 MIN

Target the rings with computer groups and Group Policy

> **WHY THIS MATTERS** Rings become mechanics: a computer group per ring on SBD-WSUS01, plus one for SBD-AUDIT01, client-side targeting so each computer files itself into its ring, and one Group Policy object per ring that names the server, the ring and its install behaviour. The domain controllers get a ring of their own: they download on their own but install only when Tier 0 starts it, and SBD-AUDIT01 waits for its audit custodian the same way.

In this lesson you will

- Create the ring groups and switch WSUS to client-side targeting.
- Build and link one Group Policy object per ring, with the pilot set by security filtering.
- Set install behaviour and deadlines, and prove every computer reports to its ring.

Follow the build

1. On SBD-WSUS01, create the four ring groups — Ring 1 - Pilot, Ring 2 - Workstations, Ring 3 - Member servers and Ring 4 - Tier 0 — and Ring 3 - Audit archive for SBD-AUDIT01 alone, and set the Computers option to Use Group Policy or registry settings on computers. The groups must exist first: a computer whose policy names a group WSUS doesn't have waits in Unassigned Computers until the group is created.
2. From AdminJumpBox01, create one GPO per ring, and one for Ring 3 - Audit archive, around the same core: Specify intranet Microsoft update service location, with https://SBD-WSUS01.sbd.test:8531 for both detection and statistics; Enable client-side targeting, with the ring's group name; and Do not connect to any Windows Update Internet locations, so no system keeps trying the public service. Leave Allow signed updates from an intranet Microsoft update service location disabled, so only Microsoft-signed updates install.
3. Set each ring's install behaviour in Configure Automatic Updates: option 4, download and schedule the install, at a time that suits the ring's work for Rings 1 to 3, and option 7 — download, then notify to install and to restart, a Windows Server option — for Ring 4, so a domain controller installs only when a Tier 0 operator starts it; Ring 3 - Audit archive uses option 7 too, so SBD-AUDIT01 installs only when its audit custodian starts it in a custodian session from SBD-ADMIN-SRV01. Deadlines go on the approvals (two days for Ring 1 and five for Rings 2 and 3 in this sample): at a deadline the update installs whatever the client's own schedule says. Ring 4's approvals carry none, and neither do Ring 3 - Audit archive's.
4. Link the GPOs where the rings live: Ring 2 at the Workstations OU; Ring 3 at the Servers and Admin Workstations OUs and at Admin Servers, filtered to SBD-ADMIN-SRV01; Ring 3 - Audit archive at the Audit Servers OU; Ring 4 at the Domain Controllers and Tier 0 Servers OUs and at Admin Servers, filtered to AdminJumpBox01. Link Ring 1 above the ring GPO on the Workstations and Servers OUs (link order 1), security-filtered to a new SBD WSUS Pilot group holding SBD-PC01–SBD-PC03 and SBD-WSUS01, and restart those four so the membership reaches their Kerberos tickets.
5. Prove the targeting on one member of each ring: run gpupdate, read the policy values, and start a detection with the Windows Update Agent's DetectNow call; then confirm on SBD-WSUS01 that each computer sits in its ring's group and that Unassigned Computers is empty. Check Ring 4 from AdminJumpBox01, and record every ring's member list against lesson 1's table.

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Check your result

```
# On SBD-WSUS01: the four ring groups and client-side targeting
$wsus = Get-WsusServer
'Ring 1 - Pilot', 'Ring 2 - Workstations', 'Ring 3 - Member servers', 'Ring 3 - Audit
archive', 'Ring 4 - Tier 0' | ForEach-Object { [void]$wsus.CreateComputerTargetGroup($_) }
$config = $wsus.GetConfiguration(); $config.TargetingMode = 'Client'; $config.Save()
# On AdminJumpBox01 (Tier 0): the links in order, and the pilot GPO's security filter
(Get-GPInheritance -Target 'OU=Workstations,OU=SBD,DC=sbd,DC=test').GpoLinks | Select-
Object Order, DisplayName
Get-GPPermission -Name 'SBD WSUS Ring 1 - Pilot' -All | Select-Object @{ n = 'Trustee'; e
= { $_.Trustee.Name } }, Permission
# On one member of each ring, after gpupdate /force: its server, its ring, and a detection
now
Get-ItemProperty HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate | Select-Object
WUServer, TargetGroup, DoNotConnectToWindowsUpdateInternetLocations
(New-Object -ComObject Microsoft.Update.AutoUpdate).DetectNow()
# Back on SBD-WSUS01: each ring's members, and nobody unassigned
Get-WsusComputer -ComputerTargetGroups 'Ring 1 - Pilot' | Select-Object FullDomainName,
LastSyncTime
Get-WsusComputer -ComputerTargetGroups 'Unassigned Computers'   # expected: no computers
```

> **WHAT SUCCESS LOOKS LIKE** The four ring groups and Ring 3 - Audit archive exist and WSUS uses client-side targeting; each ring's GPO is linked where its computers live, with the pilot filter and link order in place; one member of each ring shows SBD-WSUS01 and its ring in its policy values; every Windows computer sits in its ring's group; and Unassigned Computers is empty.

Your record

Evidence to keep: Keep the GPO reports with their links and security filtering, the pilot group's members, each ring's member list from SBD-WSUS01, the policy values from one member per ring, and the install-time and deadline table, with SBD-AUDIT01's custodian-started installs.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

A pilot PC receives both the Ring 1 and the Ring 2 GPO. Why does it still report to Ring 1?

> **REMEMBER** A group per ring, plus SBD-AUDIT01's own, and computers file themselves in by Group Policy. Rings 1–3 install on schedule and by deadline; Ring 4 waits for Tier 0, and SBD-AUDIT01 for its custodian. An empty Unassigned Computers list: every machine has a ring.

Microsoft · Step 2: Configure WSUS (TLS, computer groups, air-gapped networks)

Microsoft · Step 4: Configure Group Policy settings for automatic updates

Microsoft · Managing WSUS client computers and WSUS computer groups

Microsoft · Deploy Windows client updates using WSUS (rings and client-side targeting)

Microsoft · Get-WsusComputer

Microsoft · Group Policy scope

NIST · SP 800-53 Rev. 5 security controls

LESSON 05 / ABOUT 40 MIN

Run the first cycle and prove it

> **WHY THIS MATTERS** The design meets a real month: approve the imported updates for the pilot, promote them ring by ring on the written soak times, and install the domain controllers by hand. Then prove the result with WSUS's own reports, spot checks on the systems themselves, and the first SCAP baseline of SBD-WSUS01.

In this lesson you will

- Promote one monthly cycle through all four rings on the written timing.
- Report compliance per ring and spot-check it on the systems themselves.
- Baseline SBD-WSUS01 and hand the course's records to course 1.15.

Follow the build

1. Approve the month's updates for Ring 1 only, with its deadline, and watch the pilots install and report. Before anything moves, check that the pilot PCs still sign in and apply policy and that SBD-WSUS01 still serves its clients; then approve Ring 2 and Ring 3 on the written soak times — Ring 3 - Audit archive with Ring 3, without a deadline — recording for each promotion the numbers that justified it. SBD-AUDIT01's audit custodian then starts its install in a custodian session from SBD-ADMIN-SRV01 and records it; ordinary Tier 1 operators don't install there.
2. If an update fails or does harm, stop: leave it unapproved for the later rings and, where the update supports removal, approve it for removal in the rings that installed it. Windows client quality updates delivered through the Unified Update Platform can't be removed through WSUS, so the pilot is where they have to prove themselves.
3. Ring 4 is Tier 0 work from AdminJumpBox01. First read the approval history and confirm that every Ring 4 approval was made by a Tier 0 account — WSUS records who approved what, for which group. Then install on SBD-DC02, restart it, and check it with dcdiag and repadmin /replsummary before touching SBD-DC01; the two Tier 0 servers follow, and AdminJumpBox01 goes last.
4. Report compliance per ring with the UpdateServices cmdlets (the console's built-in reports need Microsoft's Report Viewer runtime, one more package to carry across): computers that still need or failed an approved update, and computers silent for a week. Spot-check one system per ring on the machine itself against WSUS's view of it. WSUS reports what clients tell it, so the spot checks are the independent view.
5. Close the cycle once every ring has installed: Invoke-WsusServerCleanup declines superseded updates that nothing still needs and expired ones, and removes unneeded files and computers silent for 30 days. Then take SBD-WSUS01's first SCAP baseline with course 1.4's SCC and benchmark versions, run locally in a Tier 1 session from SBD-ADMIN-SRV01, answer the IIS server and site STIG items for the WSUS web site in a STIG Viewer checklist (staged across the air gap as in course 1.4), and classify every open item as course 1.4 taught — update files on HTTP 8530 are a deviation candidate, justified by signed files whose hashes travel in TLS-protected metadata.
6. Hand off to course 1.15: the ring table, WSUS's computer list and this cycle's compliance report become inventory sources, the crossing calendar sets when its reconciliation runs, and SBD-MGMT01 joins Ring 3 simply by landing in the Servers OU. Collect the course's records into C:\Evidence\patch-protect on SBD-ADMIN-SRV01 and hash them into a manifest.

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Check your result

```
# On SBD-WSUS01 (Tier 1): approve for the pilot; deadlines are set in the console's
Approve Updates dialog
Get-WsusUpdate -Classification All -Approval Unapproved -Status FailedOrNeeded | Approve-
WsusUpdate -Action Install -TargetGroupName 'Ring 1 - Pilot'
Get-WsusComputer -ComputerTargetGroups 'Ring 1 - Pilot' -ComputerUpdateStatus
FailedOrNeeded   # expected: none once the ring is clean
Get-WsusComputer -ToLastReportedStatusTime (Get-Date).AddDays(-7)   # computers silent for
a week
# On AdminJumpBox01 (Tier 0): who approved each Ring 4 update, then each controller in
turn
$wsus = Get-WsusServer -Name SBD-WSUS01.sbd.test -PortNumber 8531 -UseSsl
$wsus.GetUpdateApprovals((New-Object Microsoft.UpdateServices.Administration.UpdateScope))
| Where-Object { $_.GetComputerTargetGroup().Name -eq 'Ring 4 - Tier 0' } | Select-Object
CreationDate, AdministratorName, Action
dcdiag /s:SBD-DC02 /q; repadmin /replsummary   # after SBD-DC02 restarts, before SBD-DC01
# Spot check from SBD-ADMIN-SRV01: what a system itself says it installed
Invoke-Command SBD-DHCP01 { Get-HotFix | Sort-Object InstalledOn -Descending | Select-
Object -First 3 HotFixID, InstalledOn }
# Back on SBD-WSUS01, once every ring has installed
Invoke-WsusServerCleanup -DeclineSupersededUpdates -DeclineExpiredUpdates -
CleanupObsoleteUpdates -CleanupUnneededContentFiles -CleanupObsoleteComputers
```

> **WHAT SUCCESS LOOKS LIKE** Each ring installed the month's updates in the written order with its promotion recorded, SBD-AUDIT01's custodian installed its updates, every Ring 4 approval names a Tier 0 account, both controllers were healthy between installs, no computer is failing or silent without an explanation, the spot checks agree with WSUS, SBD-WSUS01 has its first baseline with every open item classified, and the course's records are hashed.

Your record

Evidence to keep: Keep the per-ring approval and install timeline with the custodian's install record for SBD-AUDIT01, the Ring 4 approval history and install record, the compliance and silent-computer reports, the spot checks, the cleanup output, the SCC results and STIG Viewer checklist with classified findings, the manifest, and the handoff note to course 1.15.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

The pilot installs cleanly, but one pilot PC can no longer apply Group Policy. What happens to the promotion?

> **REMEMBER** The month moves pilot → workstations → servers → Tier 0, on record. Ring 4: approvals checked by name, controllers one at a time. Reports, spot checks and a first baseline prove it.

Microsoft · Plan your WSUS deployment

Microsoft · WSUS updates operations (approvals, deadlines, removal)

Microsoft · The WSUS Server Cleanup Wizard

Microsoft · Get-WsusComputer

Microsoft · Dcdiag

DISA · SCAP content and the SCAP Compliance Checker (SCC)

DISA · SRG/STIG tools (SCC and STIG Viewer)

NIST · SP 800-53 Rev. 5 security controls

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Design the rings and the Windows update rule	☐ yes	
02 Build SBD-WSUS01 as a disconnected server	☐ yes	
03 Carry updates across the air gap	☐ yes	
04 Target the rings with computer groups and Group Policy	☐ yes	
05 Run the first cycle and prove it	☐ yes	

Open items carried forward

Next in the learning path

That completes course 1.14. Course 1.15 builds SBD-MGMT01 and the living inventory, with this course's WSUS records among its sources.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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